SIMO NOUAYOU Celestin Gills

AI Engineer Intern — 3-month internship from August 2027 · Paris area

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SUMMARY

Engineering student at Efrei Paris (English track) focused on AI engineering. I have shipped a local LLM + OCR service in a real organisation and built products end to end: data pipelines, APIs, offline-first mobile apps, and the Docker, CI/CD and monitoring needed to deploy them.

EDUCATION

Efrei Paris — Engineering degree (Programme Grande École), 2nd year of the engineering cycle, English track *Sep 2026 – 2028*
Relevant courses: Data Analysis, Probability & Statistics, Numerical Methods, Graph Theory, Databases

Institut Saint Jean (IUSJ), Yaounde — Engineering cycle, Computer Science & Information Systems *Sep 2022 – Aug 2026*

EXPERIENCE

AI / NLP Engineering Intern · National Institute of Statistics (INS), Yaounde, Cameroon *May – Aug 2026*
On-premise AI micro-service that automates the analysis of administrative mail, integrated into the MailsoftApp application.

- **OCR pipeline:** FastAPI service handling native and scanned PDFs, images and Word files (PaddleOCR, pypdf); **617 test pages at 96.8% mean OCR confidence.**
- **NLP & LLM:** classification into **65 document types** and key-field extraction to JSON with a local LLM (Llama 3 8B via Ollama), few-shot prompting and anti-hallucination checks.
- **Delivery:** summarisation of multi-attachment mail with an async webhook mode; **400+ documents processed** in testing; fully local Docker deployment, no data sent to the cloud.

Other: Intern, Level Up Corporation (2023) · Cashier, Alimentation Las Gracias (2021) · Sales agent, SOCOCAM (2020)

PROJECTS

TRUSTY — economic identity for informal traders · team of 4, **lead developer (~90% of commits)** *Aug – Oct 2026*
Final-year project · FastAPI · PostgreSQL · React Native / Expo · Vosk · OAuth 2.0 · GitHub Actions

- Built the backend (append-only event ledger, 8 business engines) and an **explainable Trust Score** (0–100, 6 sub-indices, anti-manipulation rules) shared with banks only with the trader's consent.
- Offline-first mobile app: encrypted local ledger (SQLCipher), lossless sync, **offline voice input (Vosk)** with spoken confirmation.
- Security and operations: OWASP ASVS / MASVS review, Trivy and ZAP scans in CI, encrypted backups; release v0.9.2, ready for field pilot.

Talent–Job Matching Engine · HACKVERSE Hackathon, Polytechnique Yaounde · **backend & algorithm lead** *Apr 2026*
Python · FastAPI · NetworkX · Angular · Docker

- Skills-graph matching (Gale–Shapley) extended across 10 live challenges: **explainable recommendations, school/location bias correction, skill inference for candidates without a CV**, multilingual normalisation, offline evaluation for low-connectivity users.

SecureAI Vault — Zero-Trust secrets vault on AWS · Cloud project, pair work *Feb – May 2026*
AWS EC2 · Docker · Nginx · Prometheus · Grafana · Loki · GitHub Actions · scikit-learn

- Deployed Vaultwarden with rootless Docker, HTTPS, WAF (ModSecurity) and isolated networks; **CI/CD pipeline** building and shipping the images.
- **Monitoring** with Prometheus, Grafana, Loki and cAdvisor; Isolation Forest intrusion-detection module with e-mail alerts.

Other: medical cost prediction (Random Forest, Streamlit) · bank-marketing analytics dashboard · stock-management app for a real client (React Native, NestJS). *Repositories private; demo on request.*

HACKATHONS

WFEO World Engineering Day Hackathon 2026 (online, team of 4) — "Urban Resource Loop", plastic-waste collection hub for Yaounde. Owned the mass balance, the blockchain credit-system design and the business model (CAPEX, OPEX, ROI). Participant certificate.

SKILLS

AI & Data: machine learning, NLP, LLMs (Llama 3, Ollama, prompt engineering), OCR (PaddleOCR), anomaly detection, scikit-learn, Pandas, NumPy, NetworkX
Deployment: FastAPI, Docker / Compose, AWS (EC2, S3, ECR), GitHub Actions, Prometheus, Grafana, PostgreSQL, MySQL
Programming: Python, SQL, Java, JavaScript / TypeScript, C / C++, React Native, Angular, NestJS, Git
Languages: French (native) · English (TOEIC 780, English-taught degree) · Spanish (basic) **Interests:** football, cooking